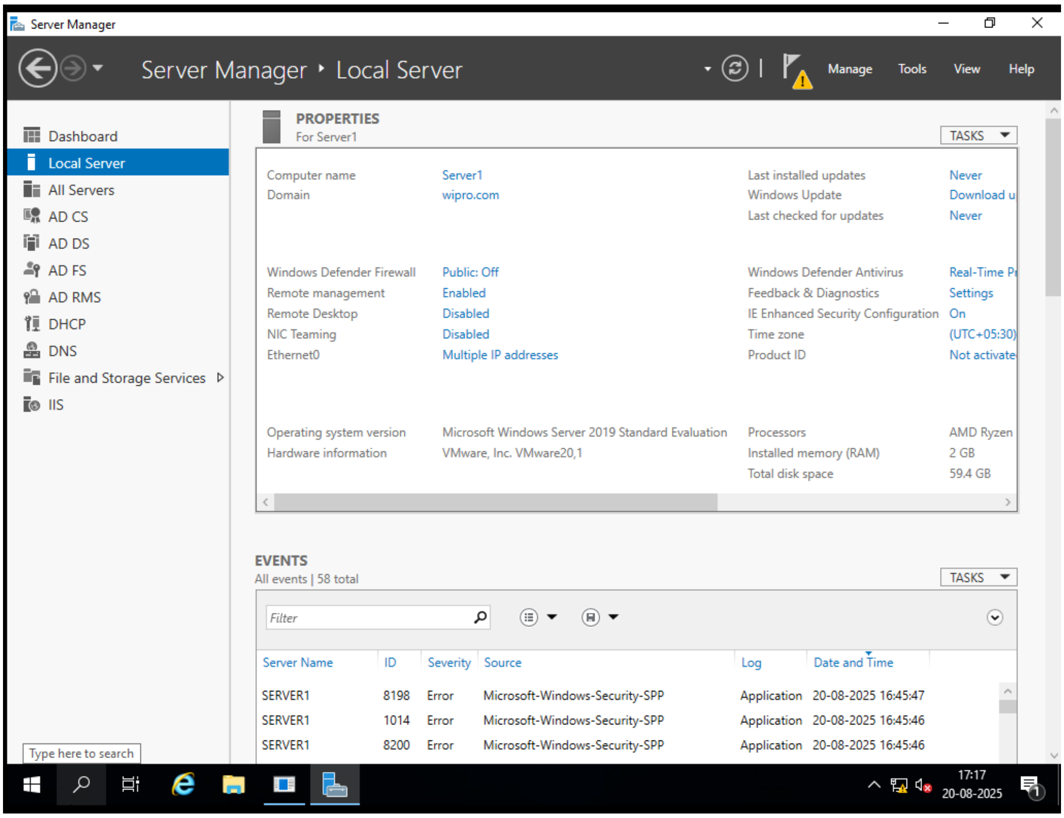
Create and Configure a Failover cluster server-2019-10

1. Prerequisites

Before creating a cluster, make sure you have:

- A **Domain Controller** (we have wipro.com).
- **Two or more Windows Server 2019 nodes** (VMs on VMware).
- Shared storage (iSCSI Target or Virtual Shared Disk).
- Static IP addresses for all cluster nodes.
- Same patch level on all cluster nodes.
- Administrative privileges.



2. Lab Setup in VMware

1. **VM1** – Domain Controller (already configured with wipro.com).
2. **VM2 & VM3** – Windows Server 2019 (Cluster Nodes).
    - Join both servers to the **wipro.com domain**.

- Assign static IPs.

3. **VM4 (Optional)** – iSCSI Target Server for shared storage.

3. Configure Shared Storage (iSCSI in VMware)

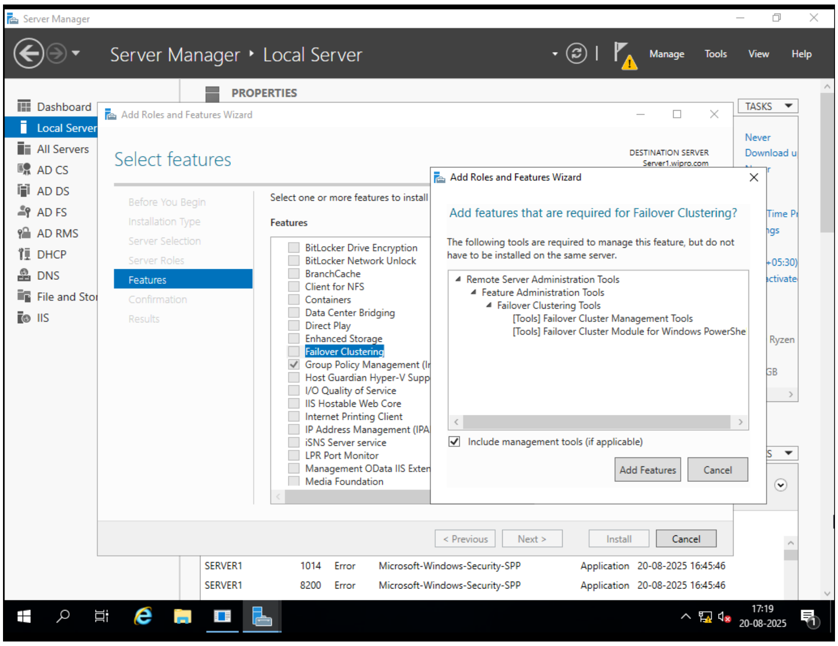
Failover Clustering requires shared storage.

1. Create a **new VM disk** in VMware and attach it to both VM2 and VM3.
   - Mark it as **shared** (in VMware Workstation: enable "Independent Persistent").
2. Initialize disk in one node, then bring it online in Disk Management.
3. Format as **NTFS** and assign a drive letter.
4. Keep this storage online only on one node at a time.

4. Install Failover Clustering Feature

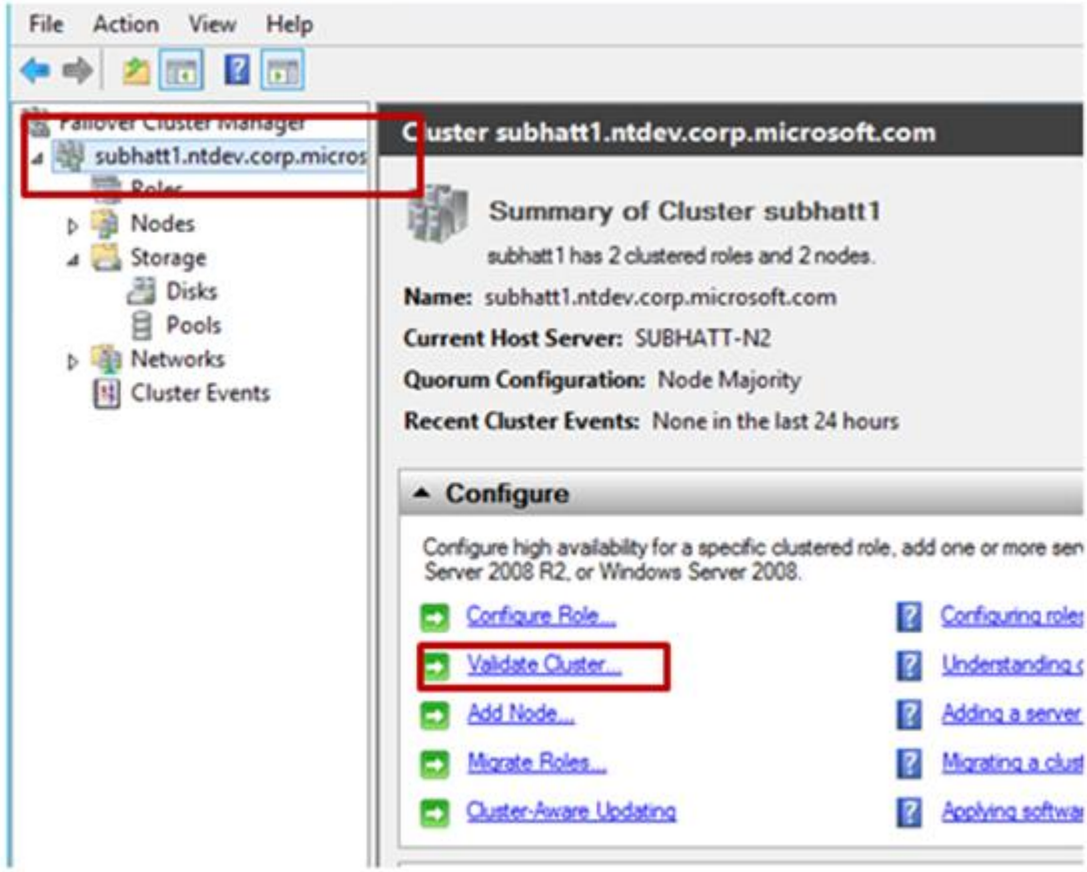
On **VM2 and VM3**:

1. Open **Server Manager → Manage → Add Roles and Features**.
2. Select **Failover Clustering** under Features.
3. Install, then restart if required.



5. Validate Cluster Configuration

1. Go to **Server Manager → Tools → Failover Cluster Manager**.
2. Select **Validate Configuration**.
3. Add both nodes (VM2.wipro.com and VM3.wipro.com).
4. Run **Validation Tests** (select all tests).
5. Ensure no critical errors.

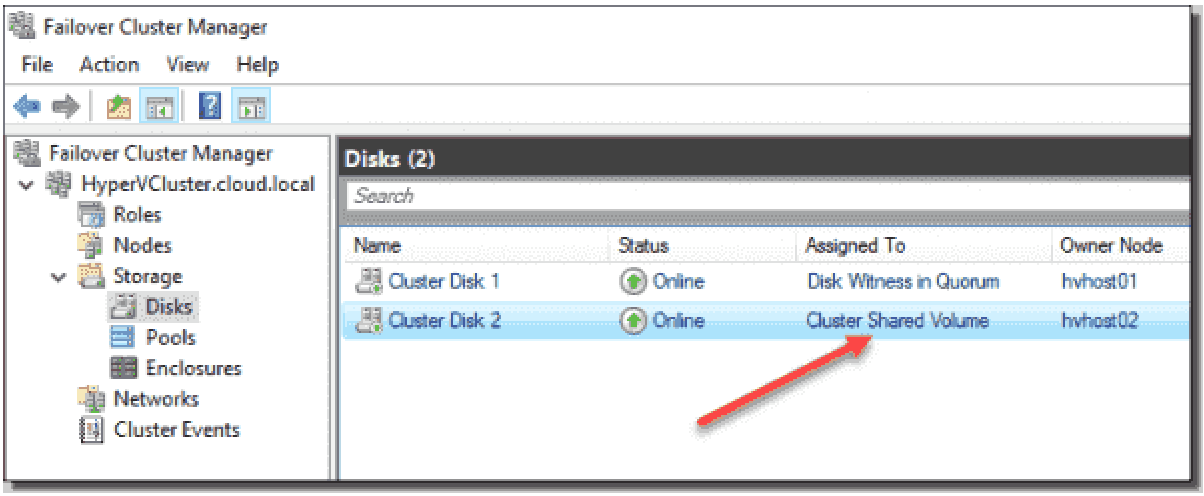


6. Create the Cluster

1. In **Failover Cluster Manager**, select **Create Cluster**.
2. Add both nodes.
3. Assign a **Cluster Name** (e.g., WiproCluster).
4. Assign a **Cluster IP address**.
5. Finish wizard, cluster will be created.

7. Configure Cluster Storage

1. In **Failover Cluster Manager**, go to **Storage → Disks**.
2. Add the shared disk created earlier.
3. Assign it to the cluster as a **Cluster Disk**.



8. Configure a Clustered Role (Example: File Server)

1. In **Failover Cluster Manager**, right-click **Roles → Configure Role**.
2. Select **File Server**.
3. Assign a **Client Access Name** (e.g., ClusterFS.wipro.com).
4. Select the cluster disk as storage.
5. Complete setup.

9. Test Failover

1. Connect from a client machine to \\ClusterFS.wipro.com.
2. Access shared folder.
3. In **Failover Cluster Manager**, move the role from Node1 to Node2.
4. Verify that the client connection is still active.

10. Verification & Benefits

- The cluster should failover between nodes without downtime.
- Benefits:

- High availability of services.
- Load balancing.
- Automatic failover.